

# tf.compat.v1.metrics.mean\_squared\_error

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[https://www.tensorflow.org/api\\_docs/python/tf/compat/v1/metrics/mean\\_squared\\_error](https://www.tensorflow.org/api_docs/python/tf/compat/v1/metrics/mean_squared_error)

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*Computes the mean squared error between the labels and predictions.*

## Function Signatures

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```
tf.compat.v1.metrics.mean_squared_error( labels, predictions,  
weights=None, metrics_collections=None,  
updates_collections=None, name=None )
```

## Code Examples

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```
tf.compat.v1.metrics.mean_squared_error(  
    labels,  
    predictions,  
    weights=None,  
    metrics_collections=None,  
    updates_collections=None,  
    name=None  
)
```

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    labels,  
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```

# Documentation

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Computes the mean squared error between the labels and predictions.

## Used in the notebooks

- Migrate metrics and optimizers

The `mean_squared_error` function creates two local variables, `total` and `count` that are used to compute the mean squared error. This average is weighted by `weights`, and it is ultimately returned as `mean_squared_error`: an idempotent operation that simply divides `total` by `count`.

For estimation of the metric over a stream of data, the function creates an `update_op` operation that updates these variables and returns the `mean_squared_error`. Internally, a `squared_error` operation computes the element-wise square of the difference between predictions and labels. Then `update_op` increments `total` with the reduced sum of the product of `weights` and `squared_error`, and it increments `count` with the reduced sum of `weights`.

If `weights` is `None`, `weights` default to 1. Use `weights` of 0 to mask values.

## Returns

## Raises